SPARKY'S SCIENCE FACTS

How Does Electricity Get to Your Home?

Electricity · Ages 7+

> **Short answer:** Electricity is made in power stations and travels to your home through long power lines.

Sparky always says: Never touch fallen power lines or fly kites near them - stay far away and tell a grown-up.

How it works

- Most power stations spin big machines called generators, which turn movement into electricity.
- In South Africa most electricity comes from burning coal, which heats water into steam to spin the generators.
- Solar panels turn sunlight straight into electricity, and wind turbines use the wind to spin generators.
- Electricity travels very far through high pylons and power lines. Transformers then lower it to a safer level for homes.

Key words: Generator: turns spinning into electricity; Renewable: energy that won't run out

Common mix-ups

- Electricity is not stored in the plug socket: it comes from the power station all the time.
- Solar panels still work on cloudy days, just not as well.

Did you know? South Africa has a lot of sunshine, so more and more homes and schools are adding solar panels.

Try it: Count the things in your home that use electricity. Which ones could run on a solar battery?

Ask Sparky: What is load-shedding? How does a solar panel work?